

Computational Replication Report:
Ding *et al.* (2021/2022),
Observation of the Orbital Rashba-Edelstein Magnetoresistance
(arXiv:2105.04495, PRL 128)

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1 Paper summary

The paper reports magnetoresistance (MR) from orbital-angular-momentum (OAM) transport — the *orbital Rashba-Edelstein magnetoresistance* (OREMR) — in a Permalloy/oxidized-Cu (Py/Cu*) heterostructure, notably *without* heavy (strong-SOC) elements. The decisive evidence is the angular dependence of the MR in three orthogonal field-rotation planes ($\alpha = xy$, $\beta = yz$, $\gamma = zx$): ordinary AMR gives a flat β scan, but the observed $\rho_{xx}(\beta) \sim \cos^2 \beta$ is SMR-like and signals an interfacial orbital Rashba-Edelstein effect. The MR ratio is thickness-independent below the oxidation depth (interfacial signature).

2 Scope statement

This is an experimental paper. We replicate the *theory framework*: the SMR/OREMR angular resistivity model, the three-plane $\cos^2$ laws, the β -scan discriminator that identifies OREMR over AMR, and the interfacial thickness signature. Measured MR magnitudes and the spin-diffusion/dephasing-length extraction require the physical samples (out of scope).

3 Model

Longitudinal SMR/OREMR resistivity (current $\parallel x$, Chen-type form):

$$\rho_{xx}(\mathbf{m}) = \rho_0 + \Delta_{\text{AMR}} m_x^2 + \Delta_{\text{SMR}} (1 - m_y^2),$$

with $\Delta_{\text{AMR}}=0.02$, $\Delta_{\text{SMR}}=0.015$ (illustrative). AMR-only sets $\Delta_{\text{SMR}}=0$.

ID	Claim	Type	Testable?	Tested?
C1	β scan flat (AMR) vs $\cos^2 \beta$ (OREMR): discriminator	theory	yes	yes
C2	Three-plane $\cos^2$ angular laws, high R^2	theory	yes	yes
C3	MR ratio interfacial (flat for $t_{\text{Cu}^*} \leq 5\text{nm}$)	theory/model	yes	yes
C4	Measured MR magnitudes; diffusion/dephasing lengths	experiment	no	out of scope

Quantity	Expected	This work	Verdict
β scan peak-to-peak (AMR only)	0 (flat)	0.0	✓
β scan peak-to-peak (OREMR)	> 0 ($\cos^2$)	0.015	✓
α scan $\cos^2$ fit R^2	$\rightarrow 1$	1.000	✓
β scan $\cos^2$ fit R^2	$\rightarrow 1$	1.000	✓
γ scan $\cos^2$ fit R^2	$\rightarrow 1$	1.000	✓
MR-ratio plateau flatness ($t \leq 5\text{nm}$)	≈ 0	0.0	✓

4 Claims addressed

5 Results vs. paper

6 Per-claim what worked / what didn't

C1 (worked). AMR-only β scan is exactly flat; adding the SMR/OREMR term produces the $\cos^2 \beta$ modulation — the paper’s decisive discriminator. **C2 (worked).** All three scans fit $\cos^2$ to $R^2 = 1$. **C3 (worked).** The MR ratio is flat below the oxidation depth (interfacial) and decays beyond. **C4 (out of scope).** Absolute MR ratios and length extractions need the measured samples.

7 Critique of the replication

Honest: C1–C3 are analytic consequences of a phenomenological model with hand-chosen coefficients. They constitute a *theoretical consistency check* that the OREMR angular-dependence framework behaves as the paper describes (and correctly distinguishes OREMR from AMR), NOT an independent reproduction of any measured number. No experimental observable is regenerated. The interfacial plateau flatness is constructed. The LLM-judge scored PARTIAL (coverage 7, agreement 6), which we adopt.

8 Verdict

PARTIAL — the SMR/OREMR angular-dependence theory, the β -scan discriminator, and the interfacial thickness signature are reproduced; measured MR magnitudes and diffusion/dephasing-length extractions are experimental and out of scope.

Open Questions

See `report/open_questions.json`: Q1 orbital vs spin mixing conductance; Q2 orbital dephasing length; Q3 OREMR/SMR/AMR separation protocol; Q4 shunting vs intrinsic OAM quenching; Q5 generality across light-metal-oxide interfaces.